Mrityunjoy Bhattacharya

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LinkedIn | Portfolio | Eligible to work in Germany (§16b AufenthG)

PROFESSIONAL SUMMARY

MSc Computational Linguistics student at the University of Stuttgart specializing in applied NLP, speech processing, and LLM-based systems, with 2.5 years of enterprise IT experience (Wipro). My Research spans from audio deepfake detection (EUSIPCO 2026), RAG pipelines, and LLM integration into existing enterprise tools; hands-on with agentic coding (Claude Code, custom agents).

TECHNICAL SKILLS

- **Programming & Databases:** Python, SQL (PostgreSQL), Java, C++
- **LLMs & AI Automation:** Claude Code, Agentic Coding, LangChain, RAG, LLM API Integration (Claude, Gemini; cloud & local), Vector Databases (Chroma), Prompt Engineering
- **Machine Learning:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, Pandas, NumPy, NLTK, OpenCV
- **NLP / Speech / Audio:** BERT, LLM Fine-tuning, Sentence Transformers, BEATs, AASIST, CQCC, Prosodic Feature Extraction, Speech-to-Text / Text-to-Speech Pipelines
- **Data Science:** Data Cleaning, Annotation, Preprocessing, EDA, Data Visualisation
- **Backend, Web & Tools:** REST APIs, WebSockets, Flask, HTML/CSS/JS, Git, Docker, MATLAB, Simulink, Splunk, ServiceNow

PUBLICATIONS

- *Beyond the Seen: Environmental Sound Deepfake Detection Across Unseen Generators.* K. Naduvathra Revi, M. Bhattacharya, S.A. Shaik Adam. **EUSIPCO 2026**.
- *Explaining Federated SPARQL Queries with How-Provenance.* Z. Asma, M. Bhattacharya, D. Hernández, L. Galárraga, G. Flouris, I. Fundulaki, K. Hose. **DKGFed Workshop, ESWC 2024**.
- *Comparative Analysis of AC-AC Power Converters for Primary Side Wireless Power Transfer System.* K. Kumar, A. De, S.R. Singh, M. Bhattacharya, J.M. Kumbhare. **i-PACT 2019**.
- *Impact of Gallium Nitride Semiconductor Devices in Tri-State Boost Converter.* K. Kumar, M. Bhattacharya, S. Garlapati, S. Banerjee. **IEEE ICSET 2019**.

KEY PROJECTS

- **Environmental Sound Deepfake Detection (ESDD) (2025):** *ICASSP 2026 Grand Challenge, Track 1; paper accepted at EUSIPCO 2026.* Proposed EAT+AASIST, combining a pre-trained Efficient Audio Transformer with a graph-based AASIST backend; achieved 2.48% EER on EnvSDD, an 81% relative improvement over the strongest baseline (13.20%), with an 88M-parameter front-end. Designed differential fine-tuning and data augmentation (codec simulation, gain normalisation) to generalise to unseen generators.
- **Audio Deepfake Detection System (2025):** Built a fusion-based deep learning model combining CQCC acoustic features with prosodic features (pitch, shimmer, HNR) to detect synthetic speech under noisy, lossy conditions; applicable to Voice Activity Detection and audio quality assessment.
- **Automated Knowledge Base & RAG Pipeline (2025):** Co-developed a LangChain RAG system automating news research: NER/sentiment features in PostgreSQL, Sentence-Transformer embeddings in Chroma, source-cited answers via Gemini 2.5 Flash, reducing LLM hallucination by grounding outputs in factual data.
- **Synthetic Data Generation with Multimodal LLMs (2025):** Built a two-stage Qwen 2.5-VL / Granite 3.2-Vision pipeline generating and labelling reasoning tasks from COCO; validated diversity via CLIP embeddings and HDBSCAN clustering; measured linguistic complexity via Flesch scores.
- **Generative AI for Music (2024):** Trained a Bidirectional RNN generating original musical sequences; built a spectrogram-based music classification system comparing STFT vs. Mel-Spectrogram inputs.

EXPERIENCE

Research Assistant (Part-time) | University of Stuttgart, TIK | Stuttgart — Jul 2026 – Present

- Integrated local and external LLM APIs into the Znuny ticketing system through custom extension packages.
- Developed German–English translation and ticket summarisation features for university agents and administrative staff.
- Collaborated with cross-functional teams to improve the user experience and streamline ticket management.

Research Assistant (Part-time) | University of Stuttgart, IABP | Stuttgart — Apr 2025 – Present

- Built a climate-monitoring system on Raspberry Pi Zero that streams readings from 5+ sensors in real time using Python.
- Ported the architecture to Arduino Giga R1 to handle multiple sensors simultaneously and cut measurement latency.

- Conducted a feasibility study on NVIDIA Jetson Nano for non-invasive temperature estimation vs. physical sensors.
- Developed an Android app (MIT App Inventor) displaying live sensor readings from IoT devices during SciWalk field sessions; contributed to the SciWalk project design connecting environmental science with hands-on IoT data collection.

Research Assistant (Part-time) | University of Stuttgart, Institute for AI (KI) | Stuttgart Jan 2024 – Sep 2026

- Built a RESTful web interface (Python, Flask) for Knowledge Graph Embeddings visualising federated SPARQL queries across distributed data sources via an asynchronous backend.
- Designed API endpoints connecting multiple heterogeneous data sources.
- Maintained and updated the AI department website to keep research findings and events current.

Project Engineer | Wipro Technologies | Kolkata, India Sep 2020 – Jan 2023

- Diagnosed and resolved a P2 live incident in BizTalk orchestration that had halted data transfer; wrote the C# fix independently, averting significant financial exposure for Royal Dutch Shell.
- Contributed to a Trading SAP system upgrade, resolving integration issues across critical SAP Chemicals interfaces; recognised with a Certificate of Appreciation by the Delivery Head.
- Reviewed and approved change requests in ServiceNow before deployment, following enterprise change management on a live production account.
- Monitored the enterprise environment for Royal Dutch Shell using Splunk and ServiceNow under SLA commitments; ran SQL queries for data verification and root-cause analysis.

Machine Learning Intern | HighRadius | Bhubaneswar, India May 2019 – Apr 2020

- Joined as a summer intern (May–Jun 2019) and continued as a Machine Learning Intern.
- Built a B2B cloud application using time-series forecasting to predict financial trends from historical data.
- Connected ML pipelines to a web front end and built interactive dashboards in HTML, CSS, and JavaScript.
- Deployed the product to the cloud with the engineering team, handling backend scaling and front-end integration for enterprise clients.

Research Intern | Indian Institute of Technology Kharagpur | Kharagpur, India May 2018 – Jun 2018

- Research internship on electric motors and electric vehicles.

EDUCATION

MSc Computational Linguistics | University of Stuttgart Oct 2024 – Present

- Specialisation: Applied NLP. Coursework: Methods of Computational Linguistics, Text Technology, Machine Translation, Deep Learning for Speech and Text Processing, Foundation Models, Machine Learning.

B.Tech Electrical Engineering | KIIT University May 2016 – May 2020

- German grade equivalent: 1.9 (German scale: 1.0 = best)

LANGUAGES

English (C1), German (A2, actively improving), Hindi (Native), Bengali (Native)